SKYG

ARDUPILOT CONFIG

SKYG SPEED 120

Matek H743-WLITE (ArduPilot firmware: MatekH743-bdshot) · ArduPlane 4.5 or later

OVERVIEW

Starting ArduPlane configuration for SkyG Speed 120 on the Matek H743-WLITE (firmware MatekH743-bdshot, ArduPlane 4.5 or later), with the rpm-governor script skyg_governor.lua. Load the .param file in Mission Planner (Config > Full Parameter List > Load from file), copy the script to the microSD card, reboot, load the .param file again (for the GOV_ parameters), then work through the bench checks.

Flight controller: on the wing-bay floor on a 0.5 mm foam pad, arrow forward; servo rail Vx set to 6 V for the ES9052MD servos.

WARNINGS

- These are starting values, not a tested setup. Verify every output, direction and endpoint on the bench before the first flight.
- Never raise GOV_RPM_MAX (38,000) or THR_MAX (70 %): they protect the prop's rpm rating.
- The servo rail (Vx) defaults to 5 V on the H743-WLITE: set it to 6 V for the ES9052MD servos (7.2 V only for the ES9052MD HV).
- No airspeed sensor: the throttle limits do not stop a dive from building speed. Fly stage runs level.

FRAME, ORIENTATION AND LOGGING

Board mounted on the wing-bay floor, arrow forward, components up (build guide). Log to the microSD card.

PARAMETER	VALUE	NOTE
AHRS_ORIENTATION	0	FC arrow forward, upright. Change only if the board is mounted otherwise.
LOG_BACKEND_TYPE	1	Logs to the microSD card.
LOG_DISARMED	0	No logging while disarmed.

OUTPUTS

S1 drives the motor with bidirectional DShot (MatekH743-bdshot firmware: outputs 1-8 bidirectional, in groups 1-2, 3-4, 5-6, 7-8): S2 stays unused. S3 drives the single aileron servo (both ailerons through the torque rods). S4 and S5 drive the left and right tail panels as a V-tail mix.

Servo endpoints are calculated for the Emax ES9052MD (about 11.1 us per degree) and the linkages: aileron 2.79:1 (12 mm horn, 4.3 mm arm hole), tail 1.19:1 (8 mm horn, 6.7 mm arm hole). They are starting points: set the final values by measuring the deflections on the bench.

PARAMETER	VALUE	NOTE
SERVO1_FUNCTION	70	S1: throttle (motor ESC).
SERVO2_FUNCTION	0	S2: unused (shares the DShot timer group with S1).
SERVO3_FUNCTION	4	S3: aileron (one servo drives both ailerons).
SERVO4_FUNCTION	79	S4: V-tail left (left tail panel).
SERVO5_FUNCTION	80	S5: V-tail right (right tail panel).
SERVO3_TRIM	1500	Aileron neutral. The 3 deg reflex is set mechanically on each link, not here (one servo moves the ailerons in opposite directions).
SERVO3_MIN	1178	About 10 deg aileron (servo 29 deg at 2.79:1). Verify: about 3.5 mm at the aileron trailing edge.
SERVO3_MAX	1822	As SERVO3_MIN. Linkage limit +-20 deg aileron = +-807 us: never set beyond it.
SERVO4_TRIM	1500	Left panel neutral; trim mechanically first (collar on the shaft flat).

OUTPUTS

PARAMETER	VALUE	NOTE
SERVO4_MIN	1232	+-20 deg panel travel at 1.19:1 (servo 24 deg). Verify by measurement.
SERVO4_MAX	1768	As SERVO4_MIN.
SERVO5_TRIM	1500	Right panel neutral.
SERVO5_MIN	1232	As SERVO4_MIN.
SERVO5_MAX	1768	As SERVO4_MIN.
MIXING_GAIN	0.5	V-tail mix: full pitch alone gives half the panel travel (about +-10 deg), pitch plus yaw can reach the +-20 deg endpoints.
SERVO_RATE	50	Servo update rate (Hz) for the PWM groups; the ES9052MD is a digital servo and also accepts higher rates.

MOTOR, ESC AND RPM TELEMETRY

The Sequre SQESC 2670 runs AM32: bidirectional DShot gives the motor rpm that the governor script uses. Check on the bench that RPM1 reads the motor rpm (Mission Planner status page) before enabling the script; if the ESC's bidirectional DShot does not work, the fallback is AM32's serial telemetry on the RX4 pad (SERIAL6_PROTOCOL 16).

SERVO_DSHOT_ESC and the protocol depend on the ESC firmware version: verify on the bench.

PARAMETER	VALUE	NOTE
SERVO_BLH_MASK	1	DShot on output 1.
SERVO_BLH_OTYPE	5	DShot300. Verify the ESC accepts it.
SERVO_BLH_BDMASK	1	Bidirectional DShot (rpm telemetry) on output 1.
SERVO_BLH_POLES	14	Motor magnet poles (12N14P, T-Motor F50 2207 2150KV). Verify.
SERVO_DSHOT_ESC	1	BLHeli_32 / AM32 type ESC. Verify against your firmware version's list.
RPM1_TYPE	5	RPM sensor from ESC telemetry (logged).
RPM1_ESC_MASK	1	Motor on output 1.

RPM GOVERNOR AND CURRENT LIMIT (LUA SCRIPT)

Copy skyg_governor.lua to APM/scripts on the microSD card. With SCR_ENABLE 1 and a reboot, the script adds its GOV_ parameters; load this file a second time to set them. The script limits the throttle so the prop stays at GOV_RPM (stage switch high) or GOV_LNCH (switch low), never above GOV_RPM_MAX (38,000 rpm; Gemfan Vortex 5x7.5E fixed electric prop: spin-test proof load: 1.2 x the 38,000 rpm governor ceiling (Gemfan publishes no rating)), and holds the battery current at GOV_I_MAX (60 A). If the rpm telemetry stops, the throttle is limited to GOV_FS_THR; if the script stops, THR_MAX takes over.

Prove it on the thrust stand before any flight: step response, overshoot <= 300 rpm, current limit, telemetry loss and script stop.

PARAMETER	VALUE	NOTE
SCR_ENABLE	1	Lua scripting on (reboot after setting).
SCR_HEAP_SIZE	100000	Script memory (bytes).
GOV_ENABLE	1	Governor on (parameter exists after the script's first run).
GOV_RPM	10400	rpm limit with the stage switch high: stage 1 (30 m/s). Raise per the stage table.
GOV_LNCH	25000	rpm limit with the stage switch low: hand launch and climb-out only (about 811 g static thrust); switch to STAGE once climbing away, before the speed builds past the stage.
GOV_SW_CH	7	Channel 7: LAUNCH / STAGE switch.
GOV_RPM_MAX	38000	Ceiling for GOV_RPM and GOV_LNCH. Never raise it.
GOV_I_MAX	60	Battery current limit (A).
GOV_KP	0.012	Limiter gain (throttle % per rpm). Tune on the thrust stand.

RPM GOVERNOR AND CURRENT LIMIT (LUA SCRIPT)

PARAMETER	VALUE	NOTE
GOV_KI	0.04	Limiter gain (throttle % per rpm-second).
GOV_KI_I	4.0	Current-limit gain (throttle % per A-second).
GOV_FS_THR	50	Throttle limit (%) when the rpm telemetry is lost.

THROTTLE

THR_MAX is a backstop: 70 % keeps an unloaded prop on a full pack at its rating if the governor script stops. The speed stages are governor rpm values (GOV_RPM, stage table), flown at full stick with the stage switch high.

Check on the bench that MANUAL is limited by the script too (it overrides the motor output in every mode).

PARAMETER	VALUE	NOTE
THR_MAX	70	backstop only: 38,000 / (2150 KV x 25.2 V) = 70 % keeps an unloaded prop on a full pack at the governor ceiling if the governor script stops. The script (GOV_ parameters) holds the prop rpm and the battery current.
THR_MIN	0	Motor stops at zero throttle in flight (gliding, flutter abort).
THR_SLEWRATE	100	Throttle slew limit (% per second).
TRIM_THROTTLE	35	Cruise throttle guess for the synthetic-airspeed modes. Set from a log of level flight.

NO AIRSPEED SENSOR

No airspeed sensor is fitted. ArduPilot uses a synthetic airspeed from GPS, wind estimate and throttle; FBWA and MANUAL do not need it.

Stall is about 13.2 m/s; AIRSPEED_MIN keeps 25 % margin. AIRSPEED_MAX matches the first speed stage and is raised with the stages.

PARAMETER	VALUE	NOTE
ARSPD_TYPE	0	No airspeed sensor.
ARSPD_USE	0	Do not use an airspeed sensor.
AIRSPEED_MIN	17	m/s; 1.25 x the 13.2 m/s stall.
AIRSPEED_CRUISE	20	m/s; cruise target for CRUISE / FBWB / RTL.
AIRSPEED_MAX	30	m/s; stage 1. Raise with each stage the test plan clears.

ATTITUDE LIMITS, PITCH RATE AND YAW DAMPER

Pitch rate is limited per speed stage so a full pull stays under the 6 g limit (table below); the file holds stage 1 (94 deg/s). Lower it before each faster stage.

The flight-dynamics model finds the Dutch roll about neutral at 13 m/s, so the yaw damper is required. Start low and tune in flight.

PARAMETER	VALUE	NOTE
PTCH_RATE_MAX	94	deg/s, stage 1 (30 m/s). See the stage table.
ROLL_LIMIT_DEG	55	Bank limit in the stabilised modes.
PTCH_LIM_MAX_DEG	25	Nose-up limit.
PTCH_LIM_MIN_DEG	-25	Nose-down limit (no steep powered dives: speed builds fast).
YAW2SRV_DAMP	0.15	Yaw damper gain, starting point. Increase in small steps if the tail wags.
KFF_RDDRMIX	0.5	Rudder (V-tail yaw) mixed with aileron for coordinated turns.

RECEIVER, MODES AND ARMING

RadioMaster RP1 V2 ELRS receiver on UART6 (TX6/RX6 pads) with CRSF, SERIAL7 with BRD_ALT_CONFIG 1. Arm with a switch on channel 5; flight modes on channel 6; channel 7 is the LAUNCH / STAGE switch for the governor.

Hand launch in FBWA with the stage switch on LAUNCH (GOV_LNCH); no auto-takeoff is needed.

PARAMETER	VALUE	NOTE
BRD_ALT_CONFIG	1	RX6 as a full UART (SERIAL7) for CRSF.
SERIAL7_PROTOCOL	23	RCIN (CRSF/ELRS) on UART6 (SERIAL7).
RC_OPTIONS	8192	Bit 13: 420k baud for ELRS.
RC5_OPTION	153	Channel 5 arms and disarms.
ARMING_RUDDER	0	No stick arming (switch only).
ARMING_CHECK	1	All pre-arm checks.
FLTMODE_CH	6	Flight-mode channel.
FLTMODE1	5	FBWA (launch, stage runs).
FLTMODE2	5	FBWA.

RECEIVER, MODES AND ARMING

PARAMETER	VALUE	NOTE
FLTMODE3	0	MANUAL.
FLTMODE4	0	MANUAL.
FLTMODE5	11	RTL.
FLTMODE6	11	RTL.

GPS AND COMPASS

TBS M10Q on UART2 (TX2/RX2 pads, SERIAL3) with its QMC5883 compass on I2C1 (DA1/CL1). The GPS sits behind the battery: its compass is 3-6 cm from the battery leads.

Either calibrate and run compass-motor interference compensation (compassmot) on the thrust stand, or disable the compass and fly on GPS heading (COMPASS_USE 0).

PARAMETER	VALUE	NOTE
SERIAL3_PROTOCOL	5	GPS on UART2 (SERIAL3).
GPS_TYPE	1	Auto (u-blox M10). Named GPS1_TYPE in ArduPlane 4.6+.
COMPASS_ENABLE	1	Use the M10Q's compass (set 0 to fly without).
COMPASS_AUTODEC	1	Automatic declination.
COMPASS_AUTO_ROT	2	Detect the compass orientation during calibration. Verify the result against the GPS's arrow.

BATTERY MONITOR AND FAILSAFES

H743-WLITE sensors: voltage divider 1:21, current 66.7 A/V (90 A continuous sense resistor). Calibrate both against a meter.

Radio failsafe: RTL after 5 s without signal. Low battery: RTL.

PARAMETER	VALUE	NOTE
BATT_MONITOR	4	Analog voltage and current.
BATT_VOLT_PIN	10	Board voltage sense.
BATT_CURR_PIN	11	Board current sense.
BATT_VOLT_MULT	21.0	1:21 divider. Calibrate against a meter.
BATT_AMP_PERVLT	66.7	Calibrate against a meter or a known charge.
BATT_CAPACITY	680	mAh (two 3S 680 in series).
BATT_LOW_VOLT	20.4	3.4 V per cell under load (6S).
BATT_LOW_MAH	170	About 25 % left.
BATT_FS_LOW_ACT	1	RTL on low battery.

BATTERY MONITOR AND FAILSAFES

PARAMETER	VALUE	NOTE
THR_FAILSAFE	1	Radio failsafe on.
FS_SHORT_ACTN	0	Short failsafe: circle (default).
FS_LONG_ACTN	1	Long failsafe: RTL.
FS_LONG_TIMEOUT	5	Seconds.

OSD

Analog OSD (AT7456E) with ground speed, throttle, current, used capacity and voltage. ArduPilot has no ground-speed alarm: watch the ground-speed number against the stage limit and judge speed from two-way passes.

PARAMETER	VALUE	NOTE
OSD_TYPE	1	Analog OSD. Use 5 (MSP DisplayPort) with the DJI O4 noses.
OSD1_ENABLE	1	Screen 1.
OSD1_GSPEED_EN	1	GPS ground speed.
OSD1_THROTTLE_EN	1	Throttle %.
OSD1_CURRENT_EN	1	Current.
OSD1_BATUSED_EN	1	Used capacity (mAh).
OSD1_BAT_VOLT_EN	1	Battery voltage.
OSD_W_BATVOLT	21.0	Battery voltage warning (V, 6S).

BENCH CHECKS BEFORE FLIGHT

- Props off. Power from a full pack; check the battery voltage and current readings against a meter.
- Control directions in MANUAL: right stick -> right aileron trailing edge up, left down. Pull -> both tail-panel trailing edges up. Right rudder -> left panel trailing edge down, right panel up (inverted V, panels below the fuselage). Fix with SERVOx_REVERSED.
- Measure deflections with a protractor: aileron about +-10 deg, tail panels +-20 deg at the endpoints; adjust SERVOx_MIN/MAX. Nothing binds or buzzes at the ends.
- FBWA: roll and pitch the airframe by hand; the surfaces must oppose the motion.
- Thrust stand (prop on, restrained): load skyg_governor.lua; logged GOVR shows rpm held at GOV_RPM with overshoot <= 300 rpm on throttle steps; the current stays at or below 60 A; unplug the rpm signal (DShot telemetry) and the throttle drops to GOV_FS_THR; stop the script (SCR_ENABLE 0, reboot) and THR_MAX 70 % keeps an unloaded prop on a full pack at or under 38,000 rpm.
- Transmitter: channel 7 LAUNCH / STAGE switch; confirm GOV_LNCH on LAUNCH and GOV_RPM on STAGE in the GOVR log.
- Radio failsafe: switch the transmitter off with the motor at low throttle (props off): motor stops or RTL behaviour as set.
- Compass: calibrate, then compassmot on the thrust stand (or COMPASS_USE 0).
- GPS: 3D fix and the OSD shows ground speed.